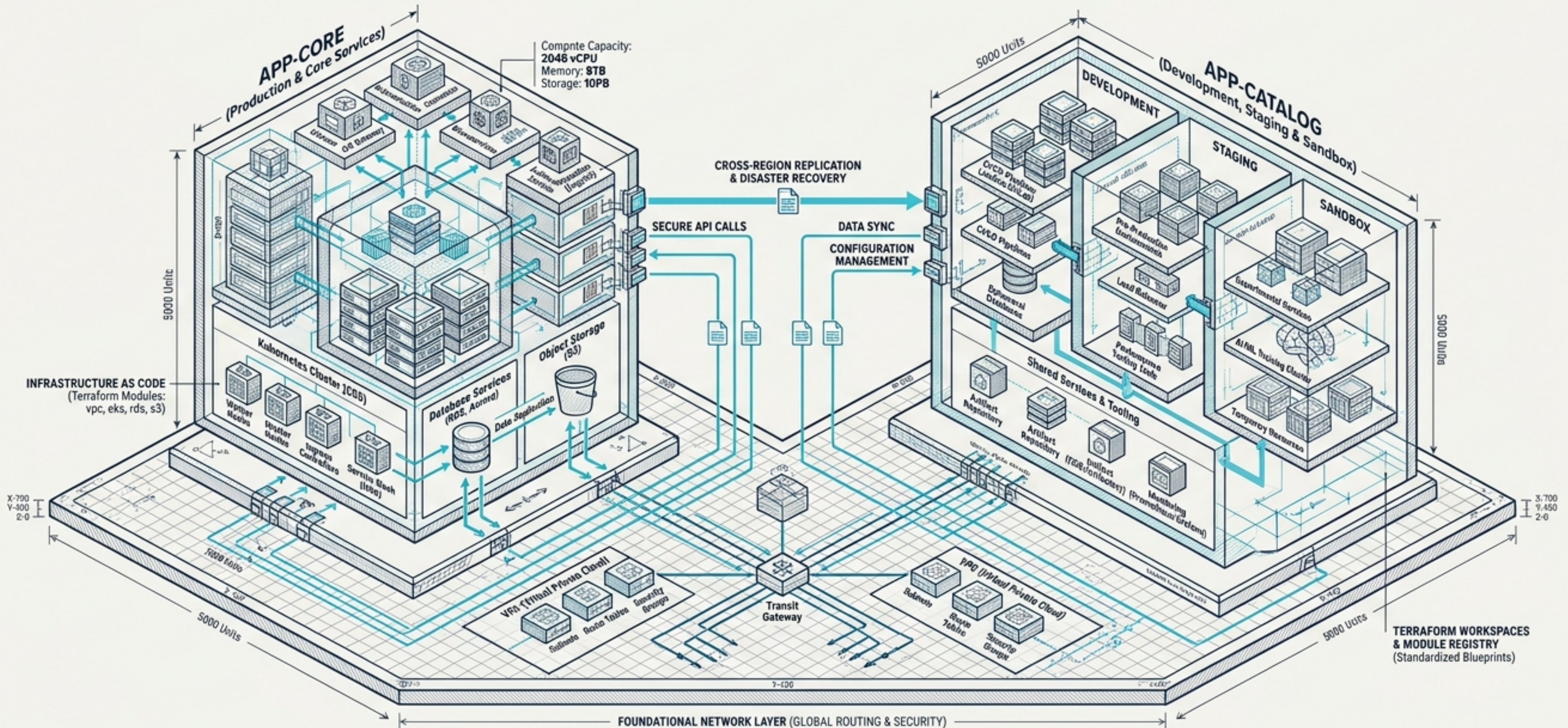


ENTERPRISE CLOUD INFRASTRUCTURE BASELINE

From High-Level Architecture to Low-Level Terraform Execution in App-Core & App-Catalog (Vision 2026)



The Enterprise Stack Hierarchy

Contextualizing Workloads within Vision 2026

Tier 4: Day2-Ops
(Observability, Ingress)

Tier 3: Business &
Catalog Workloads
(App-Core & App-Catalog)

Tier 2: AKS Hub
(Kubernetes Compute)

Tier 1: Shared-Infra
(Network Backbone, Firewalls)

Strategic Focus

Tier 3 isolates business logic and metadata from foundational networking. This decoupled architecture allows rapid application deployments without risking the core backbone.

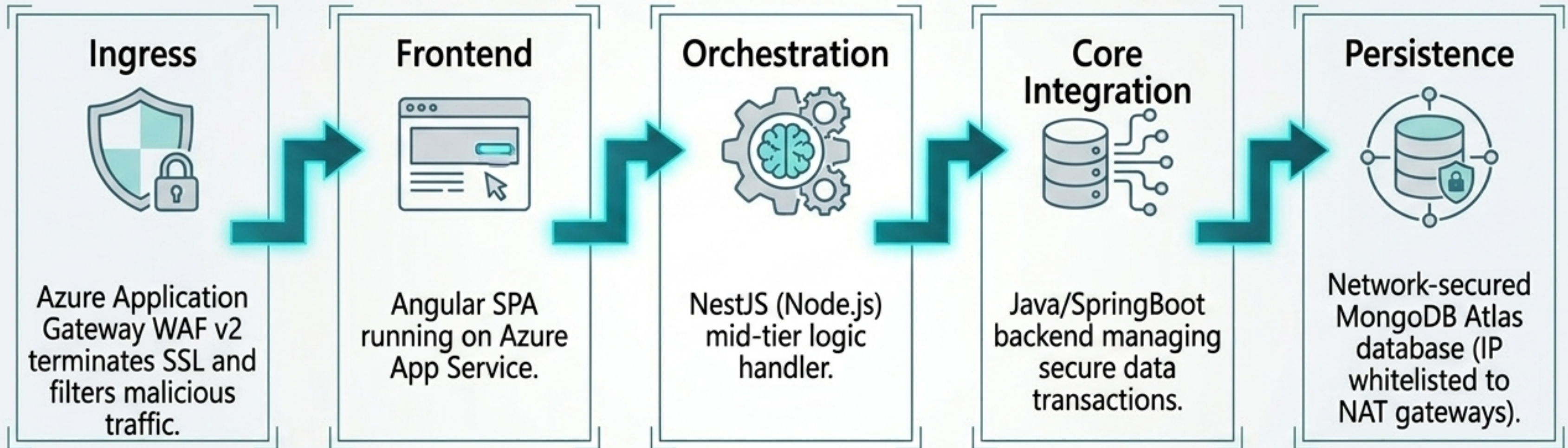
High-Level Comparison: App-Core vs. App-Catalog

		App-Core 	App-Catalog 
	Role	Business Workload	Service Registry & Metadata
	Primary Workload	Polyglot microservices (Angular, NestJS, Java) & WAF routing	Dynamic cataloging & PDF rendering for client analysis
	Key Architecture	Application Gateway + Managed APIs	Backend APIs tied to registry databases
	Terraform Footprint	High complexity (133 TF Resources)	Moderate complexity (43 TF Resources)

While App-Core manages the heavy lifting of user traffic and business logic, App-Catalog operates as the dynamic neurological registry routing metadata.

App-Core High-Level: Functional Architecture

Polyglot Ecosystem and Restricted Data Flows

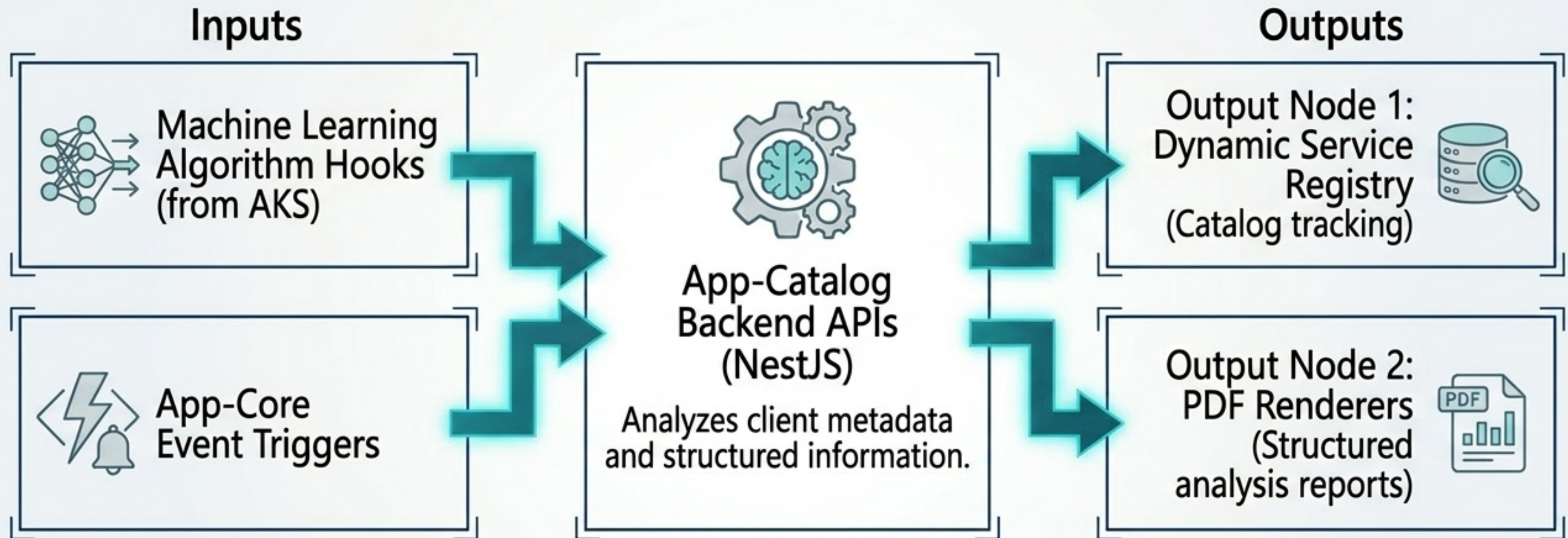


Key Insight

Data flows are restricted by design. Frontends only speak to Orchestrators; Orchestrators speak to Integrations. No direct database access from the edge.

App-Catalog High-Level: Functional Architecture

Dynamic Registry and Metadata Orchestration



App-Catalog acts as the analytical brain of the ecosystem, taking structured information from edge services and generating cataloged, readable outputs for client analysis.

Security Deep-Dive: The App-Core OBO Identity Flow

Secretless Just-In-Time (JIT) Authentication

The Edge



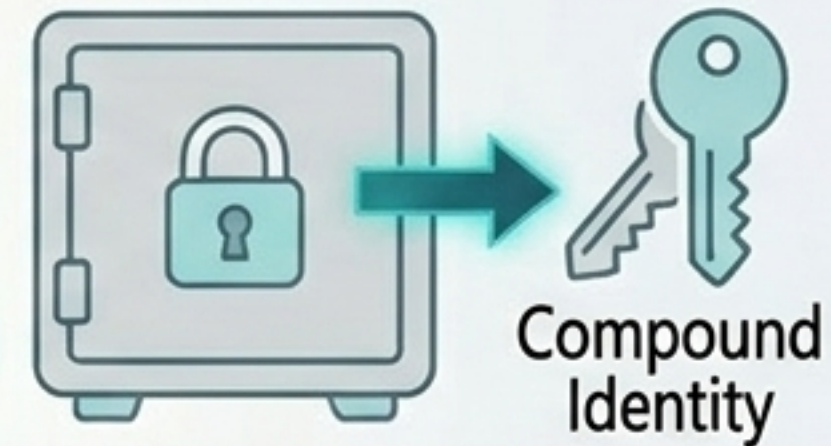
User authenticates via MSAL on the Angular SPA. Token A (ID Token) is issued by Entra ID (1hr life).

The Mid-Tier



App-Core Backend receives Token A, requests an exchange from Entra ID, receives Token B (Access Token).

The Vault



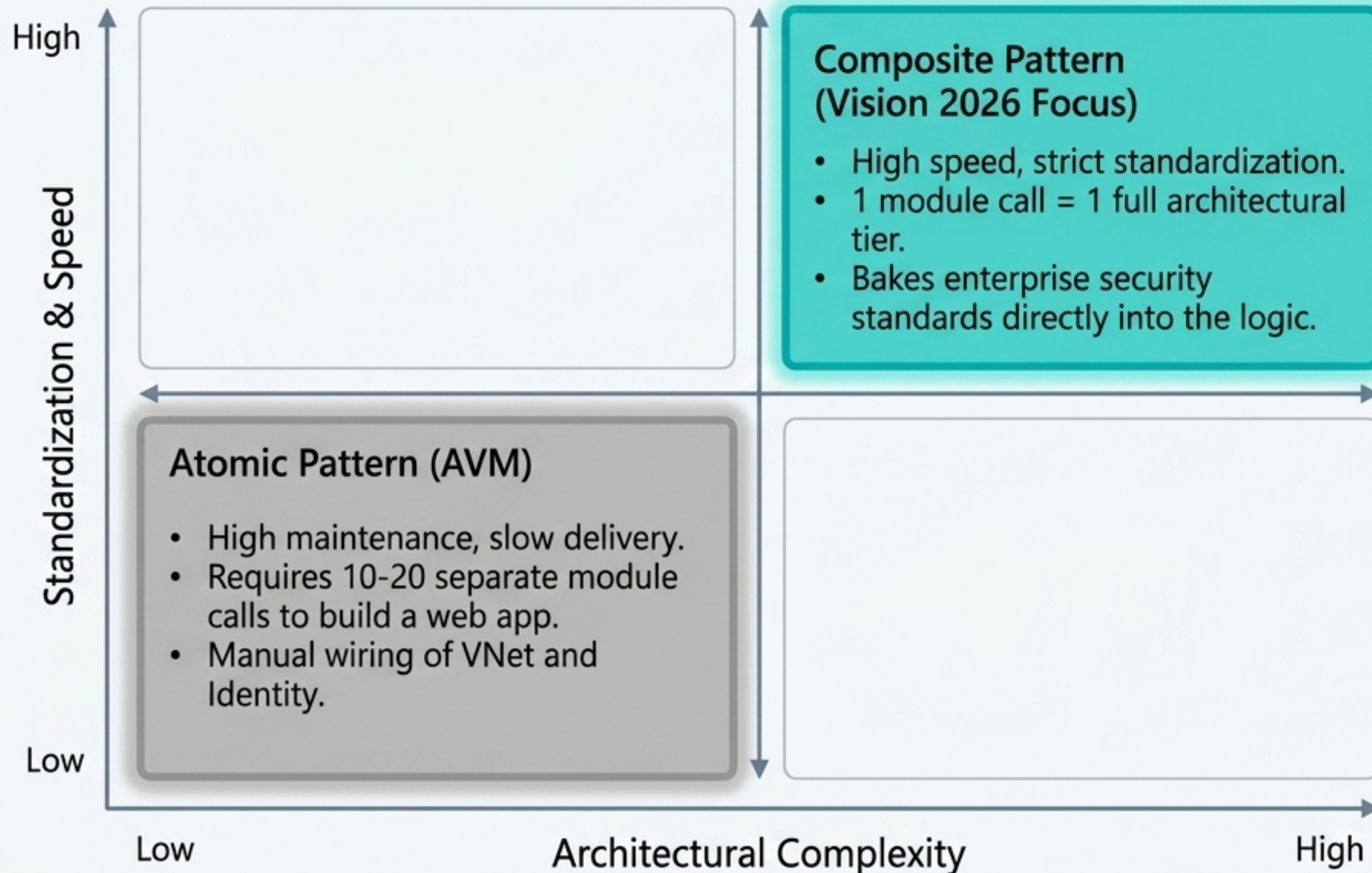
Backend utilizes the On-Behalf-Of (OBO) Flow to securely invoke Azure Key Vault using Compound Identity.

Zero-Trust Enforcement

Eradicates static credential risk. Secrets are requested Just-In-Time using short-lived tokens, ensuring downstream services know exactly WHICH human user initiated the backend call.

Terraform Strategy: Why Composite Modules?

Domain-Driven Composition vs. Atomic Sprawl

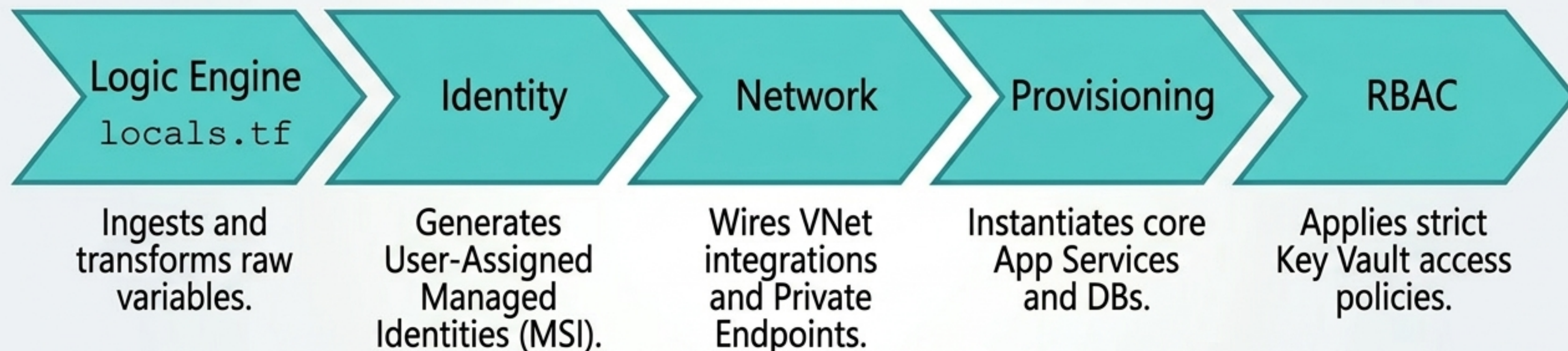


Why multiple Composite modules?

Domain-driven composition allows us to deploy an entire cohesive tier (e.g., App-Core) with a single command. It bakes enterprise security standards directly into the logic, minimizes the blast radius, and avoids the sprawl of 20+ atomic dependencies.

Low-Level IaC: The 5-Stage Execution Lifecycle

Internal Mechanics of a Composite Terraform Module



Every module in the repository strictly follows this 5-stage deterministic execution lifecycle, ensuring security is wired in BEFORE compute is provisioned.

Low-Level Code: The Custom Logic Engine

Branch Detection and Isolated Shadow Infrastructure

```
gitbranch = (var.gitbranch != 'main') ? 'd': ''
```

Path A

Input: Pipeline runs on 'develop' branch

Engine logic injects 'd' prefix

Deploys: app-back-dnedev

Isolated Shadow Infrastructure

Path B

Input: Pipeline runs on 'main' branch

Engine logic omits prefix

Deploys: app-back-nepro

Production Sovereignty

A mathematically deterministic environment generator. Engineers safely test IaC logic on the exact same codebase without ever risking production collision.

Low-Level Code: Referential Attribute Linking

Enforcing IaC Sovereignty and Eliminating Configuration Drift

[X] Hardcoded IDs / External Data

```
1 ...
2 service_plan_id =
  "/subscriptions/123/resourceGroups/rg-app
  /providers/Microsoft.Web/serverfarms/pla
3 n"
4 ...
```

[✓] Referential Attribute Linking

```
1 ...
2 service_plan_id =
  azurerm_service_plan.appcore_service_pla
  n.id
3 ...
```

IaC Sovereignty: The HCL code is the single source of truth. By passing output attributes directly as inputs to downstream resources, Terraform calculates a flawless implicit dependency graph. This eliminates hardcoded strings and prevents configuration drift.

Deployment Orchestration: Pipeline Dependencies

Mandatory Run Order for Environment Integrity

Step 1: Shared-Infra (Foundation)

Step 2: AKS (Compute Hub)

Step 3: App-Users (Identity)

Step 4: App-Core (Business)

Step 5: App-Catalog (Registry)

Decoupled State Files

Each layer maintains an isolated terraform.tfstate.

This 'Blast Radius Control' prevents a single global lock from halting the entire engineering team, while enforcing a Plan-First, Verify-Always pipeline gate.

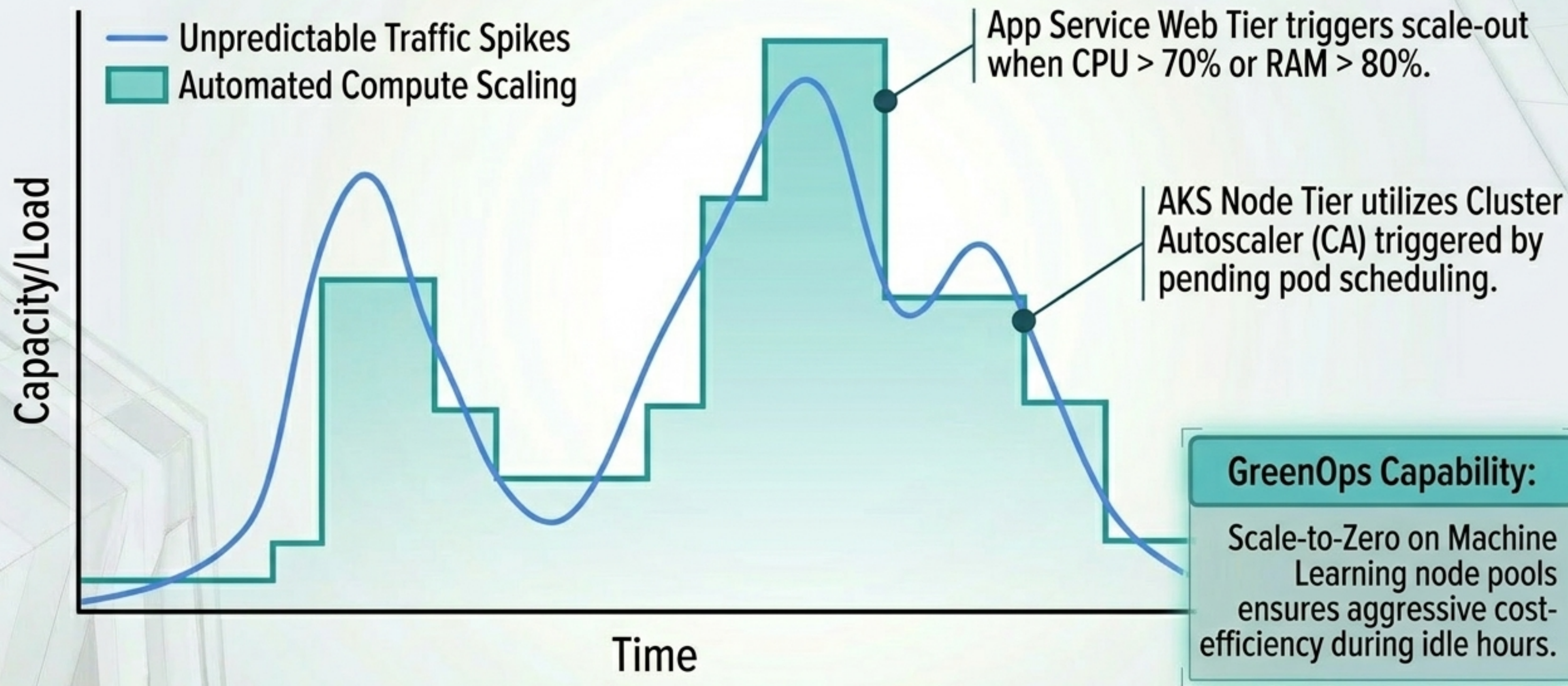
Environment Strategy: The .tfvars Matrix

Achieving Tier Parity without Leaking Secrets

	Trigger	Targets	Configuration	Zero-Secret Pipelines Credentials are NEVER stored in .tfvars files. All secrets are secrets are injected dynamically via Workload Identity Federation (OIDC) directly from Azure Key Vault at runtime.
ENG Tier	develop branch	DEV, QA, UAT, PRE	Uses eng-developbranch.tfvars	
PRO Tier	main branch	PRO, DEMO	Uses pro-mainbranch.tfvars	

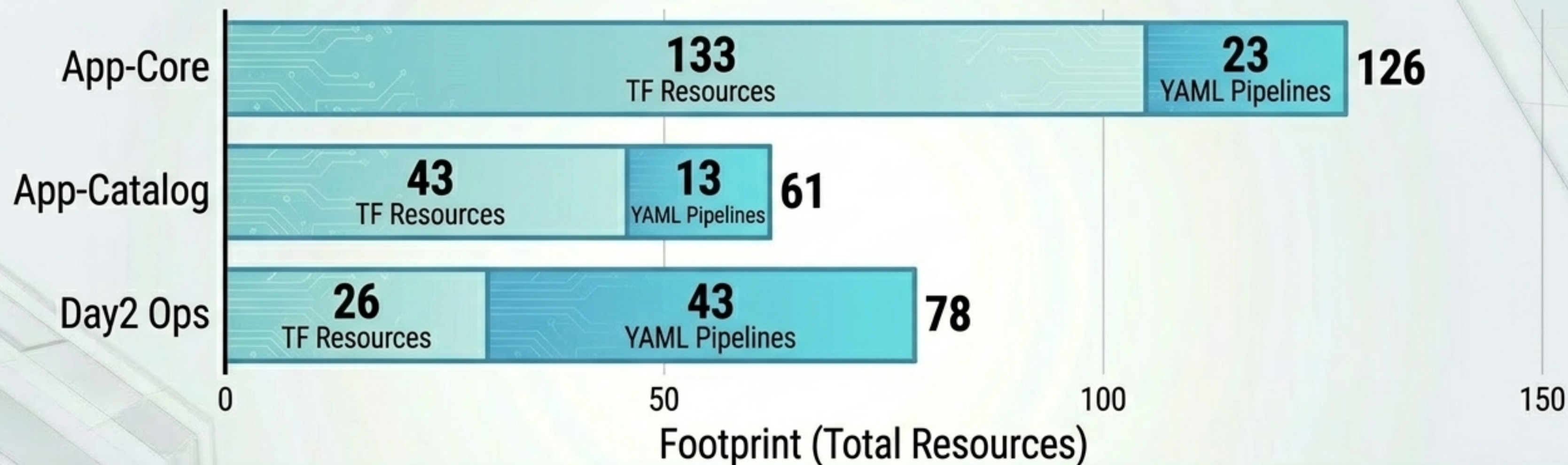
Scale & Elasticity in App-Core

Handling Enterprise Workloads Automatically



Engineering Analytics: Repository Footprint

Quantifying Domain Complexity

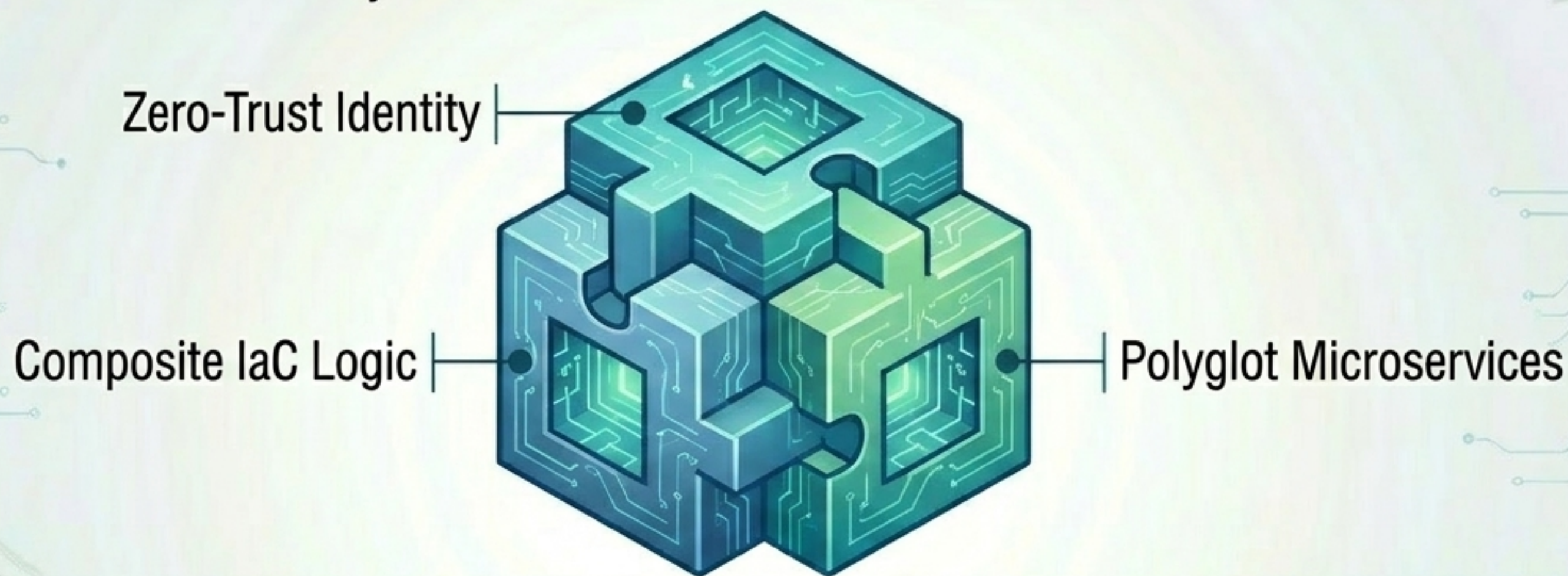


Insight

The footprint reveals intent. App-Core's heavy HCL presence reflects complex routing and logic, whereas Day2 Ops relies heavily on YAML to orchestrate Helm-based observability deployments.

Synthesis: The Immutable Enterprise Stack

Deterministic IaC Driven by Zero-Trust



“

App-Core and App-Catalog are more than just hosted applications. They are mathematically deterministic deployments. Orchestrated by composite Terraform modules, governed by dynamic .tfvars logic, and secured by OIDC identity flows, they guarantee 100% environment parity from the first line of code to the final production cloud.

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